UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis

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UAV multispectral reconstruction is challenging because multispectral bands are weak geometry drivers. They often contain lower texture and less reliable local features than RGB, while practical UAV rigs introduce cross-FOV, resolution, and alignment differences across cameras. As a result, direct spectral reconstruction can produce plausible per-band views whose 3D supports drift, biasing ratio-based products such as NDVI, GNDVI, and NDRE.

We introduce UMGS, a two-stage 3D Gaussian Splatting framework for UAV spectral-index synthesis. UMGS first reconstructs an RGB Gaussian anchor, then learns multispectral appearance while freezing Gaussian positions, covariances, opacity, and identity. This geometry lock lets low-texture spectral bands inherit stable 3D support from RGB while retaining band-specific appearance. We study two realizations: independent per-band transfer (UMGS-I) and joint band-bank transfer (UMGS-J).

Experiments on raw UAV and aligned aerial multispectral scenes show that both UMGS variants outperform comparable multispectral Gaussian baselines overall, producing stable spectral renderings and index maps while preserving shared 3D support.

CCS Concepts: • **Computing methodologies** → **Image-based rendering**; *Reconstruction*; • **Applied computing** → *Earth and atmospheric sciences.*

Additional Key Words and Phrases: multispectral imaging, 3D Gaussian Splatting, 3D reconstruction, UAV

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1 Introduction

UAV multispectral imaging is increasingly used in precision agriculture and environmental monitoring, where decisions are made from cross-band products rather than from a single rendered view [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. In this setting, reconstruction quality has two coupled requirements: (1) high per-band fidelity and (2) stable cross-band geometric correspondence. The second requirement is critical for index synthesis (e.g., NDVI, GNDVI, NDRE) because even small support drift between bands can bias ratio-based measurements [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974].

Raw UAV multispectral data make this requirement difficult to satisfy. Non-RGB bands typically provide weaker texture and less stable local features than RGB, so direct structure-from-motion or Gaussian optimization from a single spectral band is often under-constrained. At the same time, practical multispectral rigs capture RGB and G/R/RE/NIR through different cameras with different fields of view, resolutions, and spectral responses. A reconstruction method must therefore handle both weak non-RGB geometry cues and cross-camera alignment before it can produce reliable model-level products.

Recent radiance-field and Gaussian methods greatly improved training speed and view quality [Barron et al. 2022; Huang et al. 2024; Kerbl et al. 2023; Mildenhall et al. 2020; M"uller et al. 2022; Yu et al. 2024]. Emerging multispectral, hyperspectral, and thermal Gaussian lines [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025] further show that spectral rendering and index synthesis are feasible. Our focus is a narrower deployment problem: stabilizing raw UAV multispectral reconstruction when non-RGB bands provide weak geometric evidence and cross-camera misalignment directly affects downstream index synthesis.

We introduce **UMGS**, a UAV multispectral 3D Gaussian Splatting framework for spectral-index synthesis. UMGS formulates multispectral reconstruction as *RGB-anchored geometry locking*: geometry, opacity, and Gaussian identity are learned once from RGB and then frozen during G/R/RE/NIR transfer. This design turns RGB into a stable geometric carrier, while each spectral band only learns appearance on the same splat support. We evaluate two implementations of the same constraint:

- **UMGS-I**: independent per-band transfer from a shared RGB anchor.
- **UMGS-J**: joint multispectral training with band-specific appearance banks in one unified checkpoint.

Both preserve identical support by construction; they differ only in how spectral appearance parameters are organized and optimized.

For raw UAV captures, alignment quality is another bottleneck. Cross-band FOV and resolution mismatch can make naive baselines unreliable, especially for NIR. We therefore use a baseline-aware rectification QA protocol as supporting infrastructure: it separates strict pass, pass-with-warning, and fail, and logs per-frame matching/stability diagnostics rather than relying on a single opaque gate.

Our main evaluation covers a 10-scene UAV-focused benchmark: 7 raw UAV captures and 3 aligned aerial multispectral scenes. The raw track validates end-to-end applicability (alignment, QA, reconstruction, products, and geometry diagnostics). The aligned track removes rectification confounds and supports fair common-mask comparison with MS-Splatting. Across this suite, UMGS-I and UMGS-J are near-tied and consistently preserve shared support. Scratch-MS and Unfrozen-support are useful upper-bound ablations for single-band fitting, but they do not satisfy the shared-support requirement for stable multispectral index synthesis.

The contributions are:

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- A geometry-locked multispectral Gaussian formulation that uses RGB as the geometric carrier for stable spectral-index synthesis.
- Two concrete UMGS realizations implementing the same lock with independent (UMGS-I) and joint (UMGS-J) spectral parameterizations.
- A UAV-focused benchmark protocol with image, index, depth-reference, cost, and external common-mask comparisons under unified reporting.
- Supporting raw-scene infrastructure for baseline-aware rectification QA with explicit diagnostic states and decision inputs.

2 Related Work

2.1 Radiance Fields and Gaussian Scene Representations

NeRF established neural volumetric rendering as a practical paradigm for novel view synthesis [Mildenhall et al. 2019, 2020]. Follow-up work improved anti-aliasing, unbounded scenes, and training speed, including mip-NeRF/mip-NeRF360, Instant-NGP, Plenoxels, TensoRF, and DVGO [Barron et al. 2022, 2021; Chen et al. 2022; Fridovich-Keil et al. 2022; M"uller et al. 2022; Sun et al. 2022]. Large-scene and explicit-factorized lines such as NSVF, K-Planes, Zip-NeRF, MERF, Mega-NeRF, and Block-NeRF further explored memory/scale trade-offs [Barron et al. 2023; Fridovich-Keil et al. 2023; Liu et al. 2020; Reiser et al. 2023; Tancik et al. 2022; Turki et al. 2022].

3D Gaussian Splatting (3DGS) moved from volumetric integration to explicit anisotropic splats, enabling faster optimization and real-time rendering [Kerbl et al. 2023]. Subsequent work improved alias control, structural priors, compactness, and scene scale, including Mip-Splatting, Scaffold-GS, 2DGS, SuGaR, LightGaussian, Compact3DGS, Dynamic 3DGS, CityGaussian, and CityGaussianV2 [Fan et al. 2024; Guédon and Lepetit 2024; Huang et al. 2024; Lee et al. 2024; Liu et al. 2024, 2025; Lu et al. 2024b; Luiten et al. 2024; Yu et al. 2024]. These methods target rendering quality, memory, or geometry fidelity in RGB-dominant settings. UMGS keeps the standard 3DGS-style representation useful for UAV deployment, but changes how spectral appearance is attached to the geometry.

2.2 Multispectral and Thermal Gaussian Reconstruction

Multispectral remote sensing has long emphasized calibration, cross-band consistency, and physically meaningful indices [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974]. In UAV workflows, geometric preprocessing and modality alignment are often the limiting factor for downstream reliability [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. While deep learning for spectral imagery is mature in classification/segmentation tasks [Audebert et al. 2019; Li et al. 2019; Paoletti et al. 2019; Zhong et al. 2018], 3D multispectral scene reconstruction is still emerging.

Recent Gaussian-based spectral works include MS-Splatting for multi-camera multispectral NVS, HyperGS for hyperspectral latent rendering, MSGS for wavelength-aware Gaussian modeling, and thermal-domain lines such as ThermalGaussian and Thermal3D-GS [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025; Zhang et al. 2026]. These works confirm the feasibility of spectral Gaussian rendering and, in several cases, already store multiple spectral or modal signals in one Gaussian-based representation. MS-Splatting, for example, learns compact per-splat features and decodes spectral values with an MLP; HyperGS performs hyperspectral rendering in a learned latent space; ThermalGaussian learns registered RGB/thermal Gaussians after camera calibration. Our focus is complementary: raw UAV multispectral deployment, where non-RGB bands are weak geometry drivers, RGB/MS cameras can differ in FOV and resolution, and downstream products benefit from standard 3DGS-style per-band artifacts rather than only a custom neural spectral decoder.

2.3 SfM, Matching, and Cross-Modal Alignment

Practical reconstruction still depends on robust SfM/MVS foundations [Fischler and Bolles 1981; Hartley and Zisserman 2004; Sch"onberger and Frahm 2016; Sch"onberger et al. 2016]. Hand-crafted and learned feature matching both remain relevant, from SIFT/SURF/ORB and ECC-style alignment to SuperPoint, SuperGlue, LoFTR, LightGlue, and RoMa [Bay et al. 2008; DeTone et al. 2018; Edstedt et al. 2024; Evangelidis and Psarakis 2008; Lindenberger et al. 2023; Lowe 2004; Rublee et al. 2011; Sarlin et al. 2020; Sun et al. 2021]. Robust estimators (LO-RANSAC, USAC, MAGSAC) are essential under modality gaps and outlier-heavy correspondences [Barath et al. 2019; Chum et al. 2003; Raghunandan et al. 2013].

Our raw-scene QA protocol follows this systems view: it reports match quality and geometric stability separately, preserves legacy strict criteria for strong baselines, and uses a selective warning path only for weak-baseline/high-modality-gap cases. This is a protocol contribution rather than a new matcher.

2.4 Depth and Geometry Diagnostics

Depth-based reconstruction metrics are widely used in multi-view stereo and neural reconstruction [Gu et al. 2020; Wang et al. 2021; Yao et al. 2018]. However, raw UAV multispectral captures usually lack globally calibrated metric scale. We therefore evaluate geometry by relative reference-depth agreement on held-out views, and report thresholded agreement curves rather than absolute-meter accuracy claims. This positions our geometry analysis as repeatability/self-consistency under controlled probes, aligned with multispectral product reliability objectives.

3 Method

3.1 Task Definition

Given RGB and multispectral captures (G/R/RE/NIR), we seek a representation that supports both novel-view rendering and stable spectral product synthesis. Let $\mathcal{G}$ denote Gaussian support parameters (position, scale, rotation, opacity, identity) and $\mathcal{A}_b$ denote appearance parameters for band b. The key design choice in this work is to optimize $\mathcal{G}$ once from RGB, then hold $\mathcal{G}$ fixed while learning $\mathcal{A}_b$.

3.2 Stage 1: RGB Geometry Anchor

We first train an RGB Gaussian model with the standard 3DGS optimization pipeline [Kerbl et al. 2023]. RGB is used as the anchor because it provides the highest texture and most stable feature matching in our UAV captures. The resulting checkpoint defines the shared support $\mathcal{G}^\star$ used by all spectral bands.

3.3 Stage 2: Geometry-Locked Spectral Transfer

UMGS-I (independent transfer). For each band $b \in \{G, R, RE, NIR\}$, we initialize from $\mathcal{G}^\star$, freeze all support parameters, and optimize only $\mathcal{A}_b$. This yields four render-compatible spectral models sharing identical Gaussian topology by construction.

UMGS-J (joint band-bank transfer). We keep the same frozen $\mathcal{G}^\star$ but optimize four appearance banks in one joint training state. Training alternates active bands in a deterministic round-robin schedule, updates only the active bank, and writes one authoritative multispectral checkpoint. Per-band models are exported only for compatibility with existing render/metric scripts.

Representation invariants. For both UMGS-I and UMGS-J we enforce and audit:

- fixed Gaussian count and identity;
- fixed xyz/scale/rotation/opacity;
- appearance-only parameter updates in stage 2.

These invariants are the basis for index-stable shared support. A variant that optimizes per-band xyz, scale, rotation, or opacity is therefore not treated as a valid multispectral product model in our protocol, even if it improves single-band rendering. Such a model can still render each band independently, but the i-th Gaussian no longer denotes the same physical support across G/R/RE/NIR, so model-level ratios such as NDVI are no longer defined on a consistent 3D support.

3.4 Raw-Scene Rectification and Baseline-Aware QA

Raw UAV scenes require spectral-to-RGB rectification before Stage 1. We estimate per-band homographies using deterministic candidate sampling and robust aggregation from feature matches. QA is baseline-aware:

- **strict pass**: legacy edge+gradient improvement and geometry checks;
- **pass-with-warning**: allowed only for weak-baseline/high-modality-gap cases that satisfy match quality, stability, gradient gain, edge floor, and outlier-ratio constraints;
- **fail**: unstable geometry or insufficient matching evidence.

Estimator QA is an online gate; final standalone QA on exported rectified outputs is the authoritative decision.

3.5 Products and Geometry Diagnostics

We render G/R/RE/NIR and compute NDVI, GNDVI, and NDRE under a common valid mask. Geometry is evaluated by relative depth agreement on held-out probe views:

$$\delta_{rel} = \frac{D_{model} - D_{ref}}{D_{ref}}.$$

Because global metric scale is not guaranteed in raw UAV reconstructions, we report relative-depth AbsRel and threshold agreements (1%, 5%, 10%) instead of meter-based absolute accuracy.

4 Experiments

4.1 Setup

Scenes. We evaluate the main paper on a 10-scene UAV-focused benchmark: 7 raw UAV scenes (`raw_self`, `raw001–raw006`) and 3 aligned aerial multispectral scenes (`ms_golf`, `ms_lake`, `ms_solar`). Raw scenes test rectification, QA, reconstruction, products, and geometry diagnostics. Aligned scenes remove rectification confounds and are used for external common-mask comparison.

Methods. We report two UMGS variants: UMGS-I, an independent per-band transfer pipeline, and UMGS-J, a joint band-bank pipeline with one authoritative multispectral checkpoint. MS-Splatting is the external baseline on aligned UAV-style scenes. Scratch-MS and Unfrozen-support are representative single-band upper-bound ablations; they are not promoted as product models because they do not preserve the shared Gaussian support required by model-level multispectral fusion.

Metrics. Image quality: PSNR/SSIM/LPIPS. Spectral means average G/R/RE/NIR. Index fidelity: MAE/RMSE/PSNR/SSIM for NDVI, GNDVI, NDRE. Geometry: relative-depth AbsRel and agreement at 1%, 5%, 10%. Cost: non-invasive GPU traces and exported Gaussian counts. Aligned external comparison uses a common-mask protocol; UMGS-I-shim and UMGS-J-shim are resolution-aligned comparison views only, not distinct trained variants.

4.2 Main Results on UAV Scenes

Table 1 shows that UMGS-I and UMGS-J remain near-identical on aggregate quality and geometry. On the 10-scene UAV-focused suite, both reach 0.843 spectral SSIM and 0.844 index SSIM; UMGS-I has a small edge on index RMSE (0.075 vs. 0.077). Depth aggregates are identical because both variants share the same locked support.

This result supports a stable interpretation: UMGS-I remains the conservative primary line, while UMGS-J is a valid joint-checkpoint branch with numerically comparable performance.

4.3 Index-Focused Analysis

Table 2 reports NDVI/GNDVI/NDRE separately. Differences between UMGS-I and UMGS-J are small on NDVI and GNDVI, with a slightly larger NDRE gap favoring UMGS-I in RMSE/PSNR. The joint-bank design is therefore viable but not consistently superior.

4.4 Aligned External Comparison

Table 3 summarizes the aligned UAV-style common-mask comparison against MS-Splatting. The average is deliberately reported as scene-dependent: UMGS-J-shim is lower in LPIPS and has the same mean SSIM/SAM, while MS-Splatting is slightly higher in PSNR and slightly lower in spectral RMSE. Scene-level behavior in Table 4 is mixed, so we do not claim universal dominance over the external baseline.

For raw unaligned data, Table 5 is an applicability diagnostic under the external retained-subset path. It is not the main raw benchmark because it does not represent the full raw protocol used by UMGS-I/UMGS-J.

Table 1. Main UMGS results over the 10-scene UAV-focused suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR. Index metrics average NDVI/GNDVI/NDRE. Depth uses the relative reference-depth protocol and averages G/R/RE/NIR.

Method	Split	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM	AbsRel	Ag@5%	Ag@10%
UMGS-I	Raw 7	24.39	0.840	27.40	0.850	0.136	0.065	0.869	0.019	0.916	0.988
UMGS-I	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787	0.053	0.804	0.833
UMGS-I	UAV 10	24.72	0.831	27.13	0.843	0.164	0.075	0.844	0.029	0.883	0.942
UMGS-J	Raw 7	24.39	0.840	27.37	0.850	0.136	0.067	0.869	0.019	0.916	0.988
UMGS-J	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787	0.053	0.804	0.833
UMGS-J	UAV 10	24.72	0.831	27.11	0.843	0.164	0.077	0.844	0.029	0.883	0.942

Table 2. UAV-focused 10-scene spectral-index fidelity by vegetation index.

Method	Index	MAE	RMSE	PSNR	SSIM
UMGS-I	NDVI	0.058	0.083	28.49	0.847
UMGS-I	GNDVI	0.059	0.083	28.65	0.823
UMGS-I	NDRE	0.041	0.059	31.53	0.864
UMGS-J	NDVI	0.058	0.083	28.49	0.847
UMGS-J	GNDVI	0.059	0.083	28.66	0.823
UMGS-J	NDRE	0.043	0.063	31.31	0.863

Table 3. Aligned UAV-style common-mask comparison against MS-Splatting. Lower is better for LPIPS, 4-band RMSE, and SAM. UMGS shims are resolution-aligned comparison views, not separate training variants.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I-shim	27.01	0.853	0.215	0.0545	5.38
UMGS-J-shim	27.01	0.853	0.215	0.0545	5.38
MS-Splatting	27.04	0.853	0.244	0.0541	5.38

4.5 Representative Ablations

Table 6 compares the locked UMGS variants with two upper-bound ablations on `raw_self`, `raw002`, and `ms_lake`. The unfrozen branch improves selected image metrics, but it is not a valid product model: on `ms_lake`, product construction fails the shared-support audit with an xyz mismatch of 6.84×10^{-2}. Scratch-MS diverges more strongly, including Gaussian-count mismatch across bands. These failures occur at the exact stage where multispectral products are assembled, showing that per-band quality gains alone are insufficient if support consistency is lost.

4.6 Cost

Table 7 reports coarse but traceable cost indicators on the UAV-focused runs. UMGS-J uses higher recorded peak VRAM but lower average recorded VRAM in the aggregated traces; both UMGS variants retain identical Gaussian counts.

5 Limitations

First, aligned external comparison is scene-dependent rather than one-sided. Some entries favor our geometry-locked shims, while others favor MS-Splatting. Our claim is therefore reliability and protocol applicability, not universal metric dominance.

Second, UMGS-I and UMGS-J are very close numerically. This validates that the geometry-lock principle is robust to independent and joint implementations. In the current paper, UMGS-J is best interpreted as a strong alternative branch rather than a strict replacement for UMGS-I.

Third, our geometry evaluation is a reference-depth consistency protocol, not dense metric ground truth. This is appropriate for cross-band support analysis under missing global scale, but it does not replace LiDAR-grade absolute geometry benchmarks.

Fourth, cost reporting is intentionally non-invasive and currently coarse. The reported VRAM/utilization/size statistics are traceable and reproducible, but a dedicated runtime microbenchmark (e.g., standardized per-view FPS across methods and resolutions) is still future work.

6 Conclusion

We presented UMGS, a geometry-locked multispectral 3D Gaussian Splatting framework for UAV spectral-index synthesis. The method anchors geometry in RGB, then learns spectral appearance with fixed support, so cross-band products are built on consistent Gaussian topology instead of loosely coupled per-band reconstructions.

Across the UAV-focused evaluation, UMGS-I and UMGS-J achieve near-identical quality and geometry behavior, showing that the geometry-lock principle is stable under both independent and joint spectral parameterizations. On aligned UAV-style scenes, comparison with MS-Splatting is competitive and scene-dependent, while on raw scenes the full pipeline demonstrates practical applicability with rectification QA, product synthesis, and depth-reference diagnostics.

The main takeaway is methodological: for multispectral UAV reconstruction, preserving shared support is a first-order constraint, not a secondary check after optimizing single-band image metrics.

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Table 4. Scene-level aligned UAV-style common-mask comparison for UMGS-I-shim and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; 4-band RMSE and SAM evaluate the spectral vector.

Scene	UMGS PSNR	MMS PSNR	UMGS SSIM	MMS SSIM	UMGS LPIPS	MMS LPIPS	UMGS RMSE	MMS RMSE	UMGS SAM	MMS SAM
`ms_golf`	31.33	32.12	0.928	0.935	0.176	0.234	0.0300	0.0273	2.77	3.11
`ms_lake`	22.02	22.79	0.740	0.731	0.350	0.374	0.0887	0.0817	8.84	7.88
`ms_solar`	27.72	26.21	0.893	0.893	0.117	0.123	0.0448	0.0532	4.52	5.16

Table 5. Raw `self_m3m` retained-subset diagnostic. This is an applicability check for the external pipeline under its retained subset, not the main raw full-scene comparison.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I-retained	28.35	0.834	0.203	0.0502	3.98
UMGS-J-retained	28.33	0.834	0.204	0.0503	4.00
MS-Splatting-default	31.07	0.933	0.107	0.0379	3.95

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Table 6. Representative 3-scene ablation summary on `raw_self`, `raw002`, and `ms_lake`. Spectral metrics average G/R/RE/NIR; index metrics average NDVI/GNDVI/NDRE. Scratch-MS and Unfrozen-support are single-band upper-bound ablations, not valid product models, because their shared-support audits fail.

Method	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	AbsRel	Ag@5%	Shared	Products
UMGS-I	25.35	0.786	0.228	0.100	0.045	0.833	yes	yes
UMGS-J	25.33	0.786	0.228	0.100	0.045	0.833	yes	yes
Scratch-MS	25.88	0.780	0.289	0.116	0.035	0.560	no	invalid
Unfrozen-support	26.27	0.794	0.218	0.105	0.046	0.830	no	invalid

Table 7. Coarse recorded cost indicators on the UAV-focused suite. Values are aggregated from non-invasive GPU traces and exported PLY vertex counts; they are intended as a first cost table, not a microbenchmark.

Method	Peak VRAM GB	Avg. VRAM GB	Peak util. %	Gaussians M
UMGS-I	27.45	3.99	99.0	4.31
UMGS-J	35.04	1.50	99.0	4.31

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